1 A student wishes to determine the resistivity of aluminium.

For Examiner's Use

The resistivity ρ of a conductor is defined as

$$\rho = \frac{RA}{l}$$

for a conductor of resistance R, cross-sectional area A and length l.

Fig. 1.1 shows the typical dimensions of a strip of aluminium of lengths c, d and t. The resistivity of aluminium is about $10^{-8}\,\Omega$m.

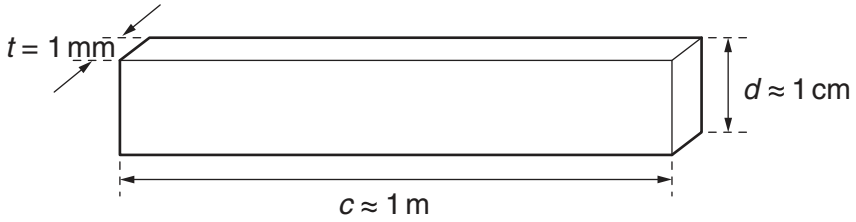

Fig. 1.1 (not to scale)

Design a laboratory experiment to determine the resistivity of aluminium using this strip. The usual apparatus of a school laboratory is available, including a metal cutter.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

[15]

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